

NZ AI Policy Sandbox – Model Equations v0.2

All 19 ANZSIC Sectors: Tiered Bounded Gain-Loss ODE System

New Zealand Whole-Economy Policy Sandbox

AI for Good New Zealand

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Abstract

This document extends the NZ AI Policy Sandbox (NZAIS) model equations from v0.1 (Tier 1, nine sectors, approximately 61% of GDP) to all 19 ANZSIC Level 1 sectors (100% of GDP). The extension enables an honest whole-economy productivity aggregate, which was the principal open flaw of v0.1.

The model adapts the bounded gain-loss ODE structure from Strauss [1] to a tiered economy. Tier 1 sectors (nine sectors) retain the full state (K_s, A_s, P_s, L_s) from v0.1 – those equations are unchanged. Tier 2 sectors (six sectors) use a reduced state (A_s, P_s) , with absorptive capability replaced by a fixed multiplier and labour pressure reported ex-post. Tier 3 sectors (four sectors) use a minimal state A_s only, with productivity reported ex-post and no labour reporting.

This v0.2 document specifies the *form* of the equations for all 19 sectors. Parameter calibration is deferred to v0.3. v0.2 is a structural extension, not a calibration exercise.

Design principles preserved from v0.1: one mechanism per layer, bounded state variables by construction, sector blocks as the unit of analysis, whole-economy denominator honesty, and comparative usefulness over predictive theatre. See `STATE-VARIABLES.md` for state variable definitions.

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1 Model Structure

1.1 Design Principle: One Mechanism Per Layer

The model follows a strict causal chain where each variable is responsible for one layer. Each arrow in the DAG below represents exactly one mechanism and appears nowhere else in the system.

$$\begin{array}{ccccccc}
 E & \rightarrow & K_s & \rightarrow & A_s & \rightarrow & P_s \rightarrow L_s \\
 & & \uparrow & & & & \\
 & & \text{feedback from } A_s & & & &
 \end{array}$$

- E does not act directly on adoption – it acts *through* K_s (enabling capacity raises absorptive capability) for Tier 1, or through the fixed absorptive multiplier β_s for Tier 2 and Tier 3.
- K_s (Tier 1) or $\beta_s E$ (Tier 2/3) does not create productivity directly – it enables adoption *through* A_s .
- A_s does not create labour pressure directly in Tier 1 – it creates productivity *through* P_s , and labour pressure emerges *through* L_s .
- P_s is driven by adoption; L_s is driven by adoption and sector structure (not by P_s directly, to avoid double counting).
- There is one feedback in Tier 1: adoption A_s reinforces the demand for capability K_s (learning-by-doing, procurement maturity). This feedback does not appear in Tier 2 or Tier 3.

No mechanism appears twice. Every causal path is traceable through a single logical chain.

1.2 Tier Structure and State Reduction

The 19 ANZSIC Level 1 sectors are organised into three tiers, each with a different level of modelling fidelity. This tiering is a modelling decision, not an empirical finding. It reflects data availability, analytical priority, and the principle of tractability over false precision.

Tier	Sectors	State	Rationale
Tier 1	9 sectors	(K_s, A_s, P_s, L_s)	Full causal chain. Evidence base sufficient for sector archetypes. Covers approximately 61% of GDP.
Tier 2	6 sectors	(A_s, P_s)	Reduced fidelity. Absorptive capability replaced by fixed multiplier. Labour reported ex-post only.
Tier 3	4 sectors	A_s	Minimal state. Included for denominator completeness. Productivity and labour are ex-post only.

The tier reductions make three structural choices, each a modelling decision:

1. **Tier 2 capability reduction.** The dynamic K_s equation is replaced by a fixed sector absorptive multiplier $\beta_s \in [0, 1]$ applied to the shared enabling stock E . This collapses the two-layer $E \rightarrow K_s \rightarrow A_s$ path into a one-layer $E \rightarrow A_s$ path for Tier 2. The structural coupling to E is preserved; the dynamic accumulation is dropped.
2. **Tier 2 labour reporting.** L_s is not a dynamic state for Tier 2. Labour adjustment pressure is reported ex-post as $L_s^{T2} = \phi_s A_s$ where $\phi_s \in [0, 1]$ is a sector labour-exposure coefficient. It does not feed back into any dynamic equation.
3. **Tier 3 state minimisation.** Only adoption A_s is dynamic. Productivity is reported ex-post as $P_s^{T3} = \psi_s A_s$ where $\psi_s \in [0, 1]$. No labour reporting. Tier 3 sectors are included solely for denominator honesty in the whole-economy aggregate.

The AGENTS-level principle that sector blocks – not firm microstates – are the unit of analysis applies at all tiers. Each tier-block is diagonal: sectors interact only through the shared $E(t)$. No sector-pair spillovers are modelled at any tier.

1.3 State Variables

Definition 1 (Tier 1 Sector State Variables). *For each sector $s \in \mathcal{S}_1$ (Tier 1, $|\mathcal{S}_1| = 9$), the model tracks four state variables, all bounded in $[0, 1]$:*

$$K_s(t) : \text{Absorptive capability – effective capacity to absorb AI into productive use} \quad (1)$$

$$A_s(t) : \text{AI adoption maturity – degree of meaningful operational AI deployment} \quad (2)$$

$$P_s(t) : \text{Realised productivity effect – normalised realisation index, bounded in } [0, 1] \quad (3)$$

$$L_s(t) : \text{Labour adjustment pressure – displacement and reconfiguration index} \quad (4)$$

Definition 2 (Tier 2 Sector State Variables). *For each sector $s \in \mathcal{S}_2$ (Tier 2, $|\mathcal{S}_2| = 6$), the model tracks two dynamic state variables:*

$$A_s(t) : \text{AI adoption maturity – bounded in } [0, 1] \quad (5)$$

$$P_s(t) : \text{Realised productivity effect – bounded in } [0, 1] \quad (6)$$

Two additional quantities are reported ex-post but are not dynamic states:

$$L_s^{T2}(t) = \phi_s A_s(t) : \text{Labour adjustment pressure (reported only)} \quad (7)$$

$$\beta_s \in [0, 1] : \text{Fixed absorptive multiplier (parameter, not a state)} \quad (8)$$

Definition 3 (Tier 3 Sector State Variables). *For each sector $s \in \mathcal{S}_3$ (Tier 3, $|\mathcal{S}_3| = 4$), the model tracks one dynamic state variable:*

$$A_s(t) : \text{AI adoption maturity – bounded in } [0, 1] \quad (9)$$

One additional quantity is reported ex-post:

$$P_s^{T3}(t) = \psi_s A_s(t) : \text{Realised productivity effect (reported only)} \quad (10)$$

$\gamma_s \in [0, 1]$ is a fixed adoption multiplier (parameter, not a state). No labour reporting for Tier 3.

Definition 4 (Economy-Wide State Variable). *There is one economy-wide enabling state, shared across all tiers:*

$$E(t) : \text{National enabling capacity – economy-wide environment for AI adoption} \quad (11)$$

bounded in $[0, 1]$.

Remark 1. $P_s(t) \in [0, 1]$ by normalisation to the sector-specific productivity ceiling $\bar{p}_s$. The raw productivity gain is $\bar{p}_s \cdot P_s(t)$, where $\bar{p}_s$ is a sector parameter calibrated separately (v0.3). In v0.2, only the normalised form is specified.

Remark 2. The GDP share ω_s for each sector is a constant input parameter, not a state variable. It is sourced from `data/sector-parameter-table.csv` (Stats NZ GDP by Industry). $\sum_{s \in \mathcal{S}} \omega_s = 1$ across all 19 sectors. See Section 6.

Tier 1 Sectors

The nine Tier 1 sectors are:

Index	Sector	Symbol
1	Agriculture	$s = \text{ag}$
2	Manufacturing	$s = \text{mfg}$
3	Professional Services	$s = \text{prof}$
4	Public Sector	$s = \text{pub}$
5	Technology	$s = \text{tech}$
6	Healthcare	$s = \text{hlth}$
7	Construction	$s = \text{con}$
8	Financial Services	$s = \text{fin}$
9	Retail & Wholesale	$s = \text{ret}$

Tier 2 Sectors

The six Tier 2 sectors are:

Index	Sector	Symbol
10	Education & Training	$s = \text{edu}$
11	Transport & Warehousing	$s = \text{trn}$
12	Accommodation & Food Services	$s = \text{acc}$
13	Administrative & Support Services	$s = \text{adm}$
14	Information Media & Telecommunications	$s = \text{imt}$
15	Utilities	$s = \text{utl}$

Tier 3 Sectors

The four Tier 3 sectors are:

Index	Sector	Symbol
16	Mining	$s = \text{min}$
17	Rental, Hiring & Real Estate	$s = \text{rnt}$
18	Arts & Recreation	$s = \text{art}$
19	Other Services	$s = \text{oth}$

2 Tier 1 Core Dynamics

Tier 1 equations are identical to v0.1. Do not modify.

Assumption 1 (Nonnegative Rates). *All rate parameters are nonnegative:*

$$\rho_K, \rho_A, \rho_P, \rho_L, \rho_E, \alpha_s, \kappa_s, \phi_s, \lambda_s, \eta_s \geq 0.$$

This ensures the bounded gain-loss forms keep all state variables in $[0, 1]$ by construction, without requiring ex-post clamping.

All equations use bounded gain-loss forms:

$$\frac{dX}{dt} = (1 - X) \cdot \rho_X \cdot f(\text{drivers}) - X \cdot \rho_X \cdot g(\text{drivers})$$

so $X \in [0, 1]$ is maintained for all $t \geq 0$. See Section 7 for the formal proposition.

2.1 Capability Stock Dynamics – $K_s(t)$

Absorptive capability grows when the national enabling environment is strong and when sector-specific support is provided. It decays in the absence of investment and maintenance.

$$\boxed{\frac{dK_s}{dt} = (1 - K_s) \cdot \rho_K \cdot (\phi_s E + G_s + \eta_s A_s) - K_s \cdot \rho_K \cdot \mu_s} \quad (12)$$

Interpretation:

- $(1 - K_s) \cdot \rho_K \cdot \phi_s E$: national enabling capacity lifts sector capability, at rate proportional to the sector's exposure ϕ_s to the national environment. For Tier 1 sectors, ϕ_s is the enabling exposure parameter; in Tier 2, the same symbol is used for a different role (labour-exposure coefficient) – see Remark after Equation (19) for disambiguation.
- $(1 - K_s) \cdot \rho_K \cdot G_s$: sector-specific policy support G_s directly builds capability.
- $(1 - K_s) \cdot \rho_K \cdot \eta_s A_s$: adoption-driven demand reinforces capability building (the one feedback in the DAG). $\eta_s \geq 0$ is the adoption feedback parameter.
- $K_s \cdot \rho_K \cdot \mu_s$: capability decays at rate μ_s (skill attrition, technology obsolescence, institutional drift).
- $(1 - K_s)$ saturation prevents $K_s > 1$; K_s decay prevents $K_s < 0$.

Remark 3. G_s is a policy control input (exogenous), not a state variable. In the base scenario (no intervention), $G_s = 0$.

2.2 AI Adoption Dynamics – $A_s(t)$

Adoption grows when capability is high and decays in the absence of enabling conditions.

$$\boxed{\frac{dA_s}{dt} = (1 - A_s) \cdot \rho_A \cdot \alpha_s K_s - A_s \cdot \rho_A \cdot (1 - K_s)} \quad (13)$$

Interpretation:

- $(1 - A_s) \cdot \rho_A \cdot \alpha_s K_s$: adoption expands when capability is available. $\alpha_s \geq 0$ is the adoption responsiveness to capability for sector s .
- $A_s \cdot \rho_A \cdot (1 - K_s)$: when capability is weak, existing adoption stalls or reverses. The $(1 - K_s)$ driver ensures that low capability creates adoption pressure loss.

Remark 4. E does not appear directly in dA_s/dt . The national enabling environment acts only through K_s . This preserves the one-mechanism-per-layer principle.

2.3 Productivity Dynamics – $P_s(t)$

Realised productivity grows as adoption matures. It decays slowly when adoption is low.

$$\boxed{\frac{dP_s}{dt} = (1 - P_s) \cdot \rho_P \cdot \kappa_s A_s - P_s \cdot \rho_P \cdot (1 - A_s)} \quad (14)$$

Interpretation:

- $(1 - P_s) \cdot \rho_P \cdot \kappa_s A_s$: realised productivity gains accumulate with adoption. $\kappa_s \geq 0$ is the productivity responsiveness to adoption for sector s .
- $P_s \cdot \rho_P \cdot (1 - A_s)$: when adoption falls, the realised gain decays back toward zero.

Remark 5. K_s does not appear directly in dP_s/dt . Capability affects productivity only through A_s . This is the one-mechanism-per-layer constraint.

2.4 Labour Adjustment Pressure – $L_s(t)$

Labour adjustment pressure rises as adoption expands, modulated by sector-specific labour sensitivity.

$$\boxed{\frac{dL_s}{dt} = (1 - L_s) \cdot \rho_L \cdot \lambda_s A_s - L_s \cdot \rho_L \cdot (1 - A_s)} \quad (15)$$

Interpretation:

- $(1 - L_s) \cdot \rho_L \cdot \lambda_s A_s$: pressure rises with adoption. $\lambda_s \geq 0$ captures sector-specific labour sensitivity.
- $L_s \cdot \rho_L \cdot (1 - A_s)$: pressure eases as adoption plateaus or reverses.

Remark 6. P_s does not appear in dL_s/dt . Labour pressure is driven by adoption, not directly by productivity gains. This avoids double counting.

2.5 National Enabling Capacity – $E(t)$

$E(t)$ is the single economy-wide state. It grows with national investment G_E and decays with depreciation.

$$\boxed{\frac{dE}{dt} = (1 - E) \cdot \rho_E \cdot G_E - E \cdot \rho_E \cdot \delta_E} \quad (16)$$

Interpretation:

- $(1 - E) \cdot \rho_E \cdot G_E$: national enabling investment G_E lifts the economy-wide enabling stock.
- $E \cdot \rho_E \cdot \delta_E$: the enabling environment depreciates at rate δ_E .

Remark 7. E acts on the whole economy simultaneously – on all 19 sectors across all three tiers. In the base scenario (no supply-side intervention), $G_E = G_E^{\text{base}} \geq 0$.

3 Tier 2 Dynamics

Tier 2 sectors use a reduced state (A_s, P_s) . The dynamic K_s equation from Tier 1 is replaced by a fixed sector absorptive multiplier $\beta_s \in [0, 1]$ applied to the economy-wide enabling stock. L_s is reported ex-post as a linear function of adoption; it is not a dynamic state.

Assumption 2 (Tier 2 Nonnegative Parameters). All Tier 2 rate parameters are nonnegative:

$$\rho_A, \rho_P \geq 0, \quad \beta_s, \kappa_s, \phi_s \in [0, 1].$$

3.1 Tier 2 Causal Structure

The Tier 2 DAG collapses the $E \rightarrow K_s \rightarrow A_s$ two-step into a single $E \rightarrow A_s$ step:

$$E \rightarrow A_s \rightarrow P_s$$

The absorptive multiplier β_s captures the sector's structural coupling to the national enabling environment. It is a fixed parameter, not a dynamic state, and does not accumulate or decay over time. This is the principal reduction from Tier 1 to Tier 2: the learning-by-doing feedback loop is collapsed out.

3.2 Tier 2 AI Adoption Dynamics – $A_s(t)$

For Tier 2 sectors, the enabling environment E acts directly on adoption, scaled by the sector absorptive multiplier β_s :

$$\boxed{\frac{dA_s}{dt} = (1 - A_s) \cdot \rho_A \cdot \beta_s E - A_s \cdot \rho_A \cdot (1 - \beta_s E)} \quad (17)$$

Interpretation:

- $(1 - A_s) \cdot \rho_A \cdot \beta_s E$: adoption expands when the enabling environment is strong and the sector has sufficient absorptive capacity ($\beta_s > 0$). The product $\beta_s E$ plays the role that $\alpha_s K_s$ plays in Tier 1.
- $A_s \cdot \rho_A \cdot (1 - \beta_s E)$: adoption stalls or reverses when the enabling environment is weak (low E) or the sector has limited absorptive capacity (low β_s).
- The $(1 - A_s)$ and A_s saturation terms bound the state in $[0, 1]$ by construction.

Remark 8. Since $\beta_s \in [0, 1]$ and $E \in [0, 1]$, the product $\beta_s E \in [0, 1]$ by construction. This ensures the loss driver $(1 - \beta_s E) \in [0, 1]$, which is required for the bounded gain-loss form to maintain $A_s \in [0, 1]$. See Proposition 1 in Section 7.

Remark 9. No sector-specific policy term G_s appears in Equation (17) in the base formulation. Policy enters Tier 2 adoption via the scenario perturbations in Section 5. This is a modelling choice: Tier 2 sectors are treated as responding to aggregate conditions rather than receiving targeted sector-specific support in the baseline.

3.3 Tier 2 Productivity Dynamics – $P_s(t)$

Tier 2 productivity follows the same form as Tier 1, driven by adoption:

$$\boxed{\frac{dP_s}{dt} = (1 - P_s) \cdot \rho_P \cdot \kappa_s A_s - P_s \cdot \rho_P \cdot (1 - A_s)} \quad (18)$$

Interpretation:

- Identical in form to Equation (14) for Tier 1. The driver is adoption A_s , scaled by sector productivity responsiveness $\kappa_s \geq 0$.
- The only structural difference from Tier 1 is that A_s here is driven by $\beta_s E$ (Equation (17)) rather than by K_s .

Remark 10. Here κ_s denotes the productivity responsiveness for a Tier 2 sector – the same role it plays for Tier 1 sectors in Equation (14). Because Tier 1 and Tier 2 sectors are disjoint, the symbol κ_s is unambiguous: for any given sector s , exactly one tier definition applies.

3.4 Tier 2 Labour Reporting (Ex-Post)

Labour adjustment pressure for Tier 2 sectors is *not* a dynamic state. It is reported ex-post as a linear function of adoption:

$$L_s^{T2}(t) = \phi_s \cdot A_s(t), \quad s \in \mathcal{S}_2 \quad (19)$$

where $\phi_s \in [0, 1]$ is the sector labour-exposure coefficient. This coefficient captures the structural labour sensitivity of the sector but does not feed back into the adoption or productivity equations. L_s^{T2} is a reporting output, not a driver of any other variable.

Remark 11. Since $\phi_s \in [0, 1]$ and $A_s \in [0, 1]$, $L_s^{T2} \in [0, 1]$ by construction.

Remark 12. In Tier 1, ϕ_s denotes the enabling exposure of sector s to the national environment (coefficient on E in Equation (12)). In Tier 2, ϕ_s denotes the sector labour-exposure coefficient in the ex-post expression (19). These are different roles for the same symbol. Because Tier 1 and Tier 2 sectors are disjoint, the symbol ϕ_s is unambiguous for any given sector s : exactly one tier definition applies. This is the same convention used for κ_s (productivity responsiveness) and α_s (adoption responsiveness) across tiers.

4 Tier 3 Dynamics

Tier 3 sectors use a minimal state: adoption A_s only. These sectors are included for denominator honesty in the whole-economy aggregate. Their contribution to the aggregate is reported through the ex-post productivity expression. No labour reporting is produced for Tier 3.

Assumption 3 (Tier 3 Nonnegative Parameters). *All Tier 3 parameters are nonnegative:*

$$\rho_A \geq 0, \quad \gamma_s, \psi_s \in [0, 1].$$

4.1 Tier 3 Causal Structure

The Tier 3 DAG is reduced to a single direct path:

$$E \rightarrow A_s$$

Productivity is reported ex-post from A_s . The adoption multiplier $\gamma_s \in [0, 1]$ captures the sector's fixed coupling to the national enabling environment, analogous to β_s in Tier 2.

4.2 Tier 3 AI Adoption Dynamics – $A_s(t)$

$$\boxed{\frac{dA_s}{dt} = (1 - A_s) \cdot \rho_A \cdot \gamma_s E - A_s \cdot \rho_A \cdot (1 - \gamma_s E)} \quad (20)$$

Interpretation:

- $(1 - A_s) \cdot \rho_A \cdot \gamma_s E$: adoption expands when the enabling environment is strong, scaled by the sector's fixed adoption multiplier γ_s .
- $A_s \cdot \rho_A \cdot (1 - \gamma_s E)$: adoption decays when the enabling environment is weak or γ_s is low.
- The structural form is identical to Tier 2 Equation (17) with γ_s replacing β_s . The two multipliers play the same structural role; separate symbols are used to allow independent calibration at v0.3.

Remark 13. Since $\gamma_s \in [0, 1]$ and $E \in [0, 1]$, the product $\gamma_s E \in [0, 1]$, so $(1 - \gamma_s E) \in [0, 1]$, and the bounded gain-loss form is satisfied. See Proposition 1.

4.3 Tier 3 Productivity Reporting (Ex-Post)

Productivity for Tier 3 sectors is *not* dynamic. It is reported ex-post as:

$$P_s^{T3}(t) = \psi_s \cdot A_s(t), \quad s \in \mathcal{S}_3 \quad (21)$$

where $\psi_s \in [0, 1]$ is the sector productivity coefficient.

Remark 14. Since $\psi_s \in [0, 1]$ and $A_s \in [0, 1]$, $P_s^{T3} \in [0, 1]$ by construction.

No labour reporting for Tier 3. These sectors are included in the model for denominator honesty only. The state reduction is explicit: any labour effects in Tier 3 sectors are not reported in this version. This is a known limit of v0.2, not an empirical claim about labour adjustment in those sectors.

5 Policy Scenarios

All three scenarios share the same equal budget $\Delta \geq 0$:

$$\Delta_A = \Delta_B = \Delta_C = \Delta.$$

This is required for like-for-like comparison. Let $n = |\mathcal{S}| = 19$ (all sectors across all tiers).

In v0.2, all three scenarios operate across all 19 sectors. The mechanism by which the budget enters the dynamics depends on the tier:

- For Tier 1 sectors, budget enters via the capability gain term in Equation (12) (adds to G_s).
- For Tier 2 sectors, budget enters via the adoption gain term in Equation (17) (adds to the gain numerator).
- For Tier 3 sectors, budget enters via the adoption gain term in Equation (20) (adds to the gain numerator).

Remark 15. The budget entry point differs by tier because the first dynamic state differs by tier. The principle is consistent: policy support enters the first dynamic state in each tier's causal chain.

5.1 Scenario A – Aggregate Policy

Policy support is distributed uniformly across all 19 sectors by equal per-sector allocation.

Perturbation: Add Δ_A/n to the relevant gain term for every sector $s \in \mathcal{S}$:

$$\begin{aligned} \text{Tier 1: } \frac{dK_s}{dt} &= (1 - K_s) \cdot \rho_K \cdot \left(\phi_s E + \frac{\Delta_A}{n} + \eta_s A_s \right) - K_s \cdot \rho_K \cdot \mu_s \\ \text{Tier 2: } \frac{dA_s}{dt} &= \left(1 - A_s \right) \cdot \rho_A \cdot \left(\beta_s E + \frac{\Delta_A}{n} \right) - A_s \cdot \rho_A \cdot (1 - \beta_s E) \\ \text{Tier 3: } \frac{dA_s}{dt} &= \left(1 - A_s \right) \cdot \rho_A \cdot \left(\gamma_s E + \frac{\Delta_A}{n} \right) - A_s \cdot \rho_A \cdot (1 - \gamma_s E) \end{aligned} \quad (22)$$

The total budget deployed is exactly $19 \times (\Delta_A/19) = \Delta_A = \Delta$.

Remark 16. Uniform allocation Δ_A/n is the natural null-hypothesis intervention: it treats all sectors as equally deserving of support regardless of adoption gap, GDP weight, or structural bottleneck. It is the benchmark against which Scenarios B and C are compared.

5.2 Scenario B – Targeted Demand-Side Policy

Policy support is weighted toward sectors with greater adoption gaps, reflecting demand-side targeting.

Perturbation: Add $\Delta_B \cdot w_s^{\text{demand}}$ to the relevant gain term, where the demand weight is:

$$w_s^{\text{demand}} = \frac{1 - A_s(0)}{\sum_{s' \in \mathcal{S}} (1 - A_{s'}(0))}, \quad \forall s \in \mathcal{S} \quad (23)$$

so that $\sum_{s \in \mathcal{S}} w_s^{\text{demand}} = 1$ and the total budget is exactly Δ_B . The perturbed equations are:

$$\begin{aligned} \text{Tier 1: } \frac{dK_s}{dt} &= (1 - K_s) \cdot \rho_K \cdot (\phi_s E + \Delta_B w_s^{\text{demand}} + \eta_s A_s) - K_s \cdot \rho_K \cdot \mu_s \\ \text{Tier 2: } \frac{dA_s}{dt} &= (1 - A_s) \cdot \rho_A \cdot (\beta_s E + \Delta_B w_s^{\text{demand}}) - A_s \cdot \rho_A \cdot (1 - \beta_s E) \\ \text{Tier 3: } \frac{dA_s}{dt} &= (1 - A_s) \cdot \rho_A \cdot (\gamma_s E + \Delta_B w_s^{\text{demand}}) - A_s \cdot \rho_A \cdot (1 - \gamma_s E) \end{aligned} \quad (24)$$

Remark 17. w_s^{demand} is evaluated at the initial condition $A_s(0)$ and held constant over the scenario horizon. This represents a one-off policy allocation decision, not a continuously adaptive rule.

Remark 18. Sectors in Tier 2 and Tier 3 with low initial adoption will receive higher demand weights under Scenario B. This includes sectors where adoption evidence is sparse; the weight reflects structural gap, not observed demand. This is a known limit.

5.3 Scenario C – Targeted Supply-Side Policy

Policy support is concentrated entirely on the national enabling stock, on the theory that lifting E creates spillover benefits across all sectors simultaneously.

Perturbation: Add Δ_C to the E gain term. No direct sector support at any tier.

$$G_E^{(C)} = G_E^{\text{base}} + \Delta_C, \quad G_s^{(C)} = 0 \quad \forall s \in \mathcal{S}. \quad (25)$$

The enabling stock equation under Scenario C:

$$\frac{dE}{dt} = (1 - E) \cdot \rho_E \cdot (G_E^{\text{base}} + \Delta_C) - E \cdot \rho_E \cdot \delta_E.$$

Remark 19. Scenario C acts on all 19 sectors through the shared $E(t)$. Tier 1 sectors benefit via the $\phi_s E$ term in Equation (12). Tier 2 sectors benefit via the $\beta_s E$ term in Equation (17). Tier 3 sectors benefit via the $\gamma_s E$ term in Equation (20). The strength of the spillover depends on ϕ_s , β_s , or γ_s as appropriate for each sector's tier.

Remark 20. The central analytical question – aggregate versus targeted, and direct support versus enabling investment – is precisely the comparison between Scenarios A, B, and C under the equal budget constraint $\Delta_A = \Delta_B = \Delta_C = \Delta$.

6 Aggregation

6.1 Whole-Economy Productivity Aggregate

In v0.2, the whole-economy productivity aggregate spans all 19 sectors:

$$\bar{P}(t) = \sum_{s \in \mathcal{S}_1} \omega_s P_s(t) + \sum_{s \in \mathcal{S}_2} \omega_s P_s(t) + \sum_{s \in \mathcal{S}_3} \omega_s \psi_s A_s(t) \quad (26)$$

where:

- $P_s(t)$ for $s \in \mathcal{S}_1$ is the dynamic productivity state from Equation (14).
- $P_s(t)$ for $s \in \mathcal{S}_2$ is the dynamic productivity state from Equation (18).
- $\psi_s A_s(t)$ for $s \in \mathcal{S}_3$ is the ex-post productivity expression from Equation (21).
- ω_s is the sector's share of whole-economy GDP, with $\sum_{s \in \mathcal{S}} \omega_s = 1$.

6.2 GDP Share Parameters

Sector GDP shares ω_s are sourced from `data/sector-parameter-table.csv` (Stats NZ GDP by Industry Dec 2025) and satisfy:

$$\sum_{s \in \mathcal{S}_1 \cup \mathcal{S}_2 \cup \mathcal{S}_3} \omega_s = 1.$$

Tier 1 covers approximately 61% of GDP; Tier 2 and Tier 3 together cover the remaining approximately 39%.

Specific ω_s values are not assigned in v0.2. Numerical parameter values, including GDP share weights, are deferred to v0.3. The functional form of the aggregate in Equation (26) is the structural commitment of v0.2.

Remark 21 (Partial Aggregate – v0.1 Historical). *The v0.1 Tier 1 partial aggregate is preserved as a reference for backward compatibility:*

$$\bar{P}_{T1}(t) = \sum_{s \in \mathcal{S}_1} \omega_s P_s(t) \quad (27)$$

This covers approximately 61% of GDP and is not the whole-economy aggregate. It remains valid as a diagnostic quantity for Tier 1 behaviour.

6.3 Uncertainty Weighting Note

The aggregate $\bar{P}(t)$ in Equation (26) is weighted by GDP share ω_s but not by modelling confidence. This is a known limit:

- Tier 1 contributions carry the highest confidence: full state equations, sector archetypes, sector-specific evidence base.
- Tier 2 contributions carry intermediate confidence: reduced state, fixed absorptive multiplier, limited sector evidence.
- Tier 3 contributions carry the lowest confidence: minimal state, ex-post productivity only, included for denominator honesty.

A GDP-weighted but confidence-unweighted aggregate may overstate certainty in the tail. This is explicitly flagged as a structural limit of v0.2 and is deferred for treatment in a later version.

7 Boundedness

Proposition 1 (State Variables Remain Bounded Across All Tiers). *Suppose all parameters satisfy Assumptions 1, 2, and 3 (nonnegative rates; $\beta_s, \gamma_s, \psi_s, \phi_s \in [0, 1]$). If all dynamic state variables satisfy their initial conditions in $[0, 1]$:*

$$K_s(0), A_s(0), P_s(0), L_s(0), E(0) \in [0, 1] \quad \forall s \in \mathcal{S}_1,$$

$$A_s(0), P_s(0) \in [0, 1] \quad \forall s \in \mathcal{S}_2,$$

$$A_s(0) \in [0, 1] \quad \forall s \in \mathcal{S}_3,$$

then for all $t \geq 0$:

$$K_s(t), A_s(t), P_s(t), L_s(t), E(t) \in [0, 1] \quad \forall s \in \mathcal{S}_1,$$

$$A_s(t), P_s(t) \in [0, 1] \quad \forall s \in \mathcal{S}_2,$$

$$A_s(t) \in [0, 1] \quad \forall s \in \mathcal{S}_3.$$

Furthermore, the ex-post quantities $L_s^{T2}(t) = \phi_s A_s(t)$, $P_s^{T3}(t) = \psi_s A_s(t)$ are also in $[0, 1]$ for all $t \geq 0$.

Proof. Tier 1. Each Tier 1 equation has the bounded gain-loss form $\dot{X} = (1 - X)f^+(X) - Xf^-(X)$ where $f^+, f^- \geq 0$ by Assumption 1. At $X = 1$: $\dot{X} = -f^-(1) \leq 0$, so X cannot increase above 1. At $X = 0$: $\dot{X} = f^+(0) \geq 0$, so X cannot decrease below 0. Hence $[0, 1]$ is positively invariant for each Tier 1 equation. (This is identical to the proof in v0.1.)

Tier 2 adoption. Equation (17): $\dot{A}_s = (1 - A_s)\rho_A\beta_s E - A_s\rho_A(1 - \beta_s E)$. Since $\beta_s \in [0, 1]$ and $E \in [0, 1]$, both $\rho_A\beta_s E \geq 0$ and $\rho_A(1 - \beta_s E) \geq 0$. The same boundary argument applies: $\dot{A}_s \leq 0$ at $A_s = 1$ and $\dot{A}_s \geq 0$ at $A_s = 0$.

Tier 2 productivity. Equation (18) has the same form as Tier 1 $\dot{P}_s$. Since $A_s \in [0, 1]$ (already established above), $\kappa_s A_s \geq 0$ and $(1 - A_s) \geq 0$. Same boundary argument applies.

Tier 3 adoption. Equation (20): same form as Equation (17) with γ_s in place of β_s . Identical argument.

Ex-post quantities. $L_s^{T2} = \phi_s A_s$: since $\phi_s \in [0, 1]$ and $A_s \in [0, 1]$, the product is in $[0, 1]$. $P_s^{T3} = \psi_s A_s$: since $\psi_s \in [0, 1]$ and $A_s \in [0, 1]$, the product is in $[0, 1]$.

Joint state space. Because each dynamic state variable in each tier is individually bounded in $[0, 1]$ and equations are coupled only through the shared $E(t) \in [0, 1]$, the joint state space $[0, 1]^{4|\mathcal{S}_1|+2|\mathcal{S}_2|+|\mathcal{S}_3|+1}$ is positively invariant. $\square$

8 Sign Correctness

8.1 Tier 1 Signs (unchanged from v0.1)

Equation	Partial derivative	Sign	Interpretation
dK_s/dt	$\partial/\partial E$	+	More enabling capacity lifts capability
dK_s/dt	$\partial/\partial G_s$	+	More sector support builds capability
dK_s/dt	$\partial/\partial A_s$	+	Adoption reinforces capability demand
dK_s/dt	$\partial/\partial K_s$	− (net)	Saturation and decay dominate at high K_s
dA_s/dt	$\partial/\partial K_s$	+	Higher capability enables more adoption
dA_s/dt	$\partial/\partial A_s$	− (net)	Saturation dominates at high A_s
dP_s/dt	$\partial/\partial A_s$	+	More adoption realises more productivity
dP_s/dt	$\partial/\partial P_s$	− (net)	Saturation dominates at high P_s
dL_s/dt	$\partial/\partial A_s$	+	More adoption raises labour pressure
dL_s/dt	$\partial/\partial L_s$	− (net)	Saturation: pressure cannot exceed 1
dE/dt	$\partial/\partial G_E$	+	More national investment lifts enabling stock
dE/dt	$\partial/\partial E$	− (net)	Saturation and depreciation dominate at high E

8.2 Tier 2 Signs

Equation	Partial derivative	Sign	Interpretation
dA_s/dt	$\partial/\partial E$	+	Stronger enabling environment lifts adoption
dA_s/dt	$\partial/\partial \beta_s$	+	Higher absorptive capacity lifts adoption
dA_s/dt	$\partial/\partial A_s$	– (net)	Saturation dominates at high A_s
dP_s/dt	$\partial/\partial A_s$	+	More adoption realises more productivity
dP_s/dt	$\partial/\partial P_s$	– (net)	Saturation dominates at high P_s
L_s^{T2}	$\partial/\partial A_s$	+	Labour pressure rises with adoption

8.3 Tier 3 Signs

Equation	Partial derivative	Sign	Interpretation
dA_s/dt	$\partial/\partial E$	+	Stronger enabling environment lifts adoption
dA_s/dt	$\partial/\partial \gamma_s$	+	Higher adoption multiplier lifts adoption
dA_s/dt	$\partial/\partial A_s$	– (net)	Saturation dominates at high A_s
P_s^{T3}	$\partial/\partial A_s$	+	Productivity rises proportionally with adoption

All signs are economically sensible and consistent with the causal DAGs for each tier.

9 Calibration Pointers

Version 0.2 specifies the *form* of the equations for all 19 sectors. Numerical parameter values are deferred to v0.3. This section notes what each parameter class requires.

Source: All baseline parameter values will be drawn from `data/sector-parameter-table.csv` (348-cell parameter table, 35% pre-populated as of v0.1 release). Evidence quality is classified as observed, derived, or assumed per the data philosophy in `METHODS.md`.

Tier 1 Parameters (unchanged from v0.1)

Parameter	Description	Evidence class
ω_s	Sector GDP share	Observed (Stats NZ)
$A_s(0)$	Initial adoption state	Derived / assumed (sector research)
$K_s(0)$	Initial capability state	Assumed (digital maturity proxies)
$P_s(0)$	Initial productivity state	Assumed (sector productivity evidence)
$L_s(0)$	Initial labour pressure	Assumed (automation exposure logic)
$E(0)$	Initial enabling stock	Assumed (NZ digital infrastructure signals)
$\rho_K, \rho_A, \rho_P, \rho_L, \rho_E$	Adjustment speeds	Assumed (tuned for plausibility)
α_s	Adoption responsiveness to K_s	Derived (sector archetype)
κ_s	Productivity responsiveness	Derived (OECD productivity range)
λ_s	Labour sensitivity	Derived (automation exposure)
ϕ_s	Enabling exposure	Assumed (sector digital openness)
η_s	Adoption feedback to K_s	Assumed (learning-by-doing proxy)
μ_s	Capability decay rate	Assumed
δ_E	Enabling stock decay	Assumed

Tier 2 Parameters (new in v0.2)

Parameter	Description	Evidence class
ω_s	Sector GDP share	Observed (Stats NZ)
$A_s(0)$	Initial adoption state	Assumed (limited sector evidence)
$P_s(0)$	Initial productivity state	Assumed
β_s	Absorptive multiplier (replaces K_s)	Assumed (sector digital maturity proxy)
κ_s	Productivity responsiveness to adoption	Assumed (OECD range, Tier 2 sectors)
ϕ_s	Labour-exposure coefficient	Assumed (automation exposure, Tier 2)
ρ_A, ρ_P	Adjustment speeds	Assumed (shared with Tier 1 or sector-specific)

Tier 3 Parameters (new in v0.2)

Parameter	Description	Evidence class
ω_s	Sector GDP share	Observed (Stats NZ)
$A_s(0)$	Initial adoption state	Assumed (very limited evidence)
γ_s	Adoption multiplier	Assumed
ψ_s	Productivity coefficient	Assumed
ρ_A	Adoption adjustment speed	Assumed

Remark 22. *Do not interpret any specific numerical values in v0.2 as calibrated. Parameter values shown in any illustrative computations are placeholders. The v0.3 calibration note will provide sector-specific fitted values anchored to the evidence in `data/sector-parameter-table.csv`.*

10 Known Limits of v0.2

This document is explicit about what it does and does not claim.

1. **Parameter values not calibrated – deferred to v0.3.** Version 0.2 specifies the functional form of all 19 sector equations. No sector-specific parameter values have been fitted to data. All parameters (β_s , γ_s , ψ_s , ϕ_s , initial conditions, adjustment speeds) remain at placeholder status. Calibration is v0.3 work.
2. **Tier 2 and Tier 3 state reductions are modelling choices, not empirical findings.** The decision to collapse Tier 2 capability into a fixed multiplier β_s , to omit dynamic L_s for Tier 2, and to reduce Tier 3 to A_s only are structural simplifications chosen for tractability. They are not supported by sector-specific evidence that these reductions are valid. The reductions trade fidelity for tractability; the tradeoff is explicit, not hidden.
3. **No sector-pair spillovers.** Each sector block remains diagonal – sectors interact only through the shared enabling stock $E(t)$. Direct sector-pair linkages (e.g. Technology enabling Manufacturing, Financial Services intermediating Construction) are not modelled at any tier. This is the principal structural simplification preserved from v0.1.
4. **Deterministic ODE only.** This is a deterministic, continuous-time ODE system. No stochastic shocks, no uncertainty propagation, no Monte Carlo. Stochastic extensions are deferred to a later version.
5. **Single-country model.** The model represents New Zealand only. There are no trade terms, international spillovers, or cross-border adoption dynamics. NZ’s small-open-economy characteristics are not yet modelled.
6. **Tier 2 and Tier 3 labour effects are reported-only, not dynamic.** $L_s^{T2} = \phi_s A_s$ for Tier 2 sectors and no labour reporting for Tier 3 are observation outputs, not feedback

channels. They cannot influence policy response in any scenario comparison. Any conclusion about labour adjustment for Tier 2 or Tier 3 sectors based on this model would be extrapolation beyond the model’s current scope.

7. **Aggregate uncertainty weighting not implemented.** The whole-economy aggregate $\bar{P}(t)$ in Equation (26) is weighted by GDP share ω_s but not by modelling confidence. Tier 3 contributions enter $\bar{P}(t)$ with the same GDP-proportional weight as Tier 1 contributions, despite carrying substantially more model uncertainty per unit of weight. This is a known structural limit flagged for treatment in a later version.
8. **Policy controls are constant within a scenario.** G_s and G_E are treated as fixed scenario parameters, not adaptive policies that respond to system state. An adaptive or feedback policy formulation is deferred.

11 References

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